

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

33rd National Mathematics Olympiad - September – 2001

Screening Test – Junior Level (IX and X Std)

Note:

1. Attempt all the questions.
 2. For each question choose the most appropriate answer
 3. Write the answers in the response sheet supplied separately
 4. You may use a rough sheet, if necessary, to do calculation.
 5. Calculators/Log tables are not to be used
 6. In case of doubts regarding the mode of answering ask your supervisor for clarification.
 7. **The time allotted for answering is 2 hours only.**
 8. Figures in the questions are not drawn to scale.
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If a, b, c are real numbers such that

$$a + (1/b) = 7/3, b + (1/c) = 4, c + (1/a) = 1 \text{ then } abc =$$

- A. 2 B. 1 C. $7/3$ D. $2/3$

If $\{ (3(230 + x))^2 = 492a04 \}$ then the value of a is

- A. 4 B. 3 C. 8 D. 2

If $200 \leq a \leq 400$ and $600 \leq b \leq 1600$, then the largest possible value of the rational number b/a is

- A. 3 B. 8 C. $3/2$ D. $11/4$

If $2^{1998} - 2^{1999} - 2^{2000} + 2^{2001} = k \cdot 2^{2002}$ then the value of k is

- A. $16/3$ B. $3/16$ C. $8/3$ D. $3/8$

The number of Pythagorean triples (a, b, c) where $a^2 + b^2 = c^2$ and each integer a, b, c being odd is

- A. 3 B. 1 C. 0 D. infinite

In the year 2002, Great Britain will host the International Mathematics Olympiad. Let I, M, O be distinct positive integers such that the product $I \cdot M \cdot O = 2002$. Then the largest possible value of the sum $I + M + O$ is

- A. 1004 B. 158 C. 2004 D. 114

8. If the Number $N = 30a0b03$ is divisible by 13, then a cannot take the value.

A. 3

B. 2

C. 1

D. 0

9. If a, b, c are real numbers such that $(2a-3)^2 + (4b-5)^2 + (6c-7)^2 = 0$

A. $abc = a + b + c$

B. $abc > a + b + c$

C. $abc < a + b + c$

D. $abc = \sqrt{a + b + c}$

10. If the numbers 2^{2001} and 5^{2001} are written one after another (in decimal notation) then the total number of digits written altogether is

A. 2001

B. 2002

C. 7^{2001}

D. 10^{2001}

11. Consider the series $1-4+7-10+13-\dots$ where the $(2n)^{\text{th}}$ term is $-2(3n-1)$ and the $(2n+1)^{\text{th}}$ term is $(6n+1)$. The sum to first 2001 terms is

A. 2001

B. 3001

C. 304001

D. -301001

12. There are 20 people around a table. Each of them shakes hands with the people to his or her immediate right and immediate left. The number of handshakes that take place is

A. 20

B. 40

C. 41

D. 111

13. All odd numbers from 1 to 99 (both inclusive) are multiplied together. The two right most digits of the product is

A. 05

B. 25

C. 45

D. 65

14. f is a real function such that $f(x+y) = f(xy)$ and $f(-2/3) = -2/3$ for all x, y . Then $f(2001)$ is

A. 2001

B. $-2001/3$

C. $2001/3$

D. $2/3$

15. Between 5 p.m. and 6 p.m., I looked at my watch. Mistaking the hour hand for the minute hand, I took the time to be 57 minutes earlier than the correct time. The correct time actually was

A. 5.03

B. 5.58

C. 5.24

D. 5.22

16. In a school whose student strength is 500, two-thirds of the students who do not wear spectacles do not bring lunch. Three-quarters of the students who do not bring lunch do not wear spectacles. On the whole, 60 students wear spectacles and also bring lunch. The number of students who do not wear spectacles and who do not bring lunch is

17. The number of real roots of the equation

$$2x^{2001} + 3x^{2000} + 2x^{1999} + 3x^{1998} + \dots + 2x + 3 = 0 \text{ is}$$

A. 1

B. 2

C. 2001

D. Infinite

18. In solving a quadratic equation, one student copied the equation wrongly making an error only in the constant term and obtained 8, 2 as roots; another student also copied the equation wrongly making an error only in the coefficient of the first degree term and obtained -9, -1 as roots. The correct equation was

A. $x^2 - 10x + 9 = 0$

B. $x^2 + 10x + 9 = 0$

C. $x^2 - 10x + 16 = 0$

D. $x^2 + 10x - 16 = 0$

19. If $x \leq y \leq z$ are positive integers such that $(1/x) + (1/y) + (1/z) = 7/15$ then x cannot be

A. 7

B. 3

C. 4

D. 5

20. Men and women devotees totalling 12, men outnumbering women, visited a Swamiji and sought his blessings. The Swamiji distributed 142 flowers. Each man got same number of flowers, while each woman got two more than each man. How many flowers did each woman get?

A. 13

B. 11

C. 12

D. 9

21. For a class, copies of 9 Maths books and 16 Science books cost Rs.220/-. Each book costs an integral number of rupees. Then the cost of each Maths book is

A. Rs. 6

B. Rs. 5

C. Rs.12

D. Rs. 7

22. The real line $y = mx + c$ is such that $mc > 0$. Then the point which cannot lie on this line is

A. (0,2001)

B. (0, -2001)

C. (19,99)

D. (2001,0)

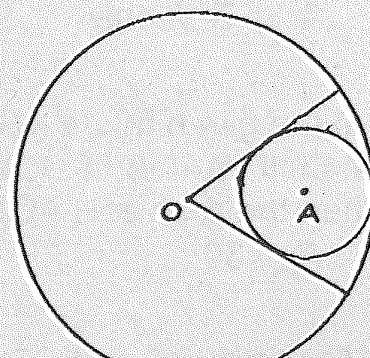
23. Circle A is inscribed in a 60° sector of circle O as shown. Then the ratio of the length of the radius of circle O to the radius of circle A is

A. 2:1

B. $\sqrt{3}:2$

C. 3:2

D. 3:1



24. A ball was floating in a lake in Himalayas, when the lake froze. The ball was removed (without breaking the ice), leaving a hole 24 cm across at the top and 8 cm deep. The radius of the ball (in cm) was

- A. 13 B. $8\sqrt{3}$ C. $6\sqrt{6}$ D. 12

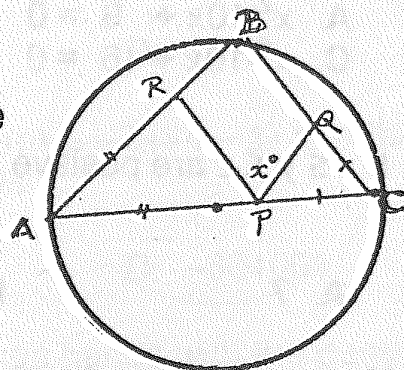
25. The radius of a circle inscribed in a rhombus whose diagonals are of length 16 and 30 is

- A. 14 B. $480/17$ C. $240/17$ D. 17

26. In the figure AC is a diameter of the circle.

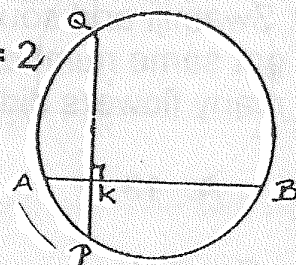
Q and R are points on BC and AC respectively, such that $PC = QC$ and $PA = RA$. Then the measure of the angle QPR is

- A. 30° B. 45°
C. 60° D. 90°



27. In the circle shown, chords AB and PQ meet at K and are perpendicular to one another. If $AK = 4$, $KB = 6$ and $PK = 2$, then the area of the circle is

- A. 45π B. 48π
C. 50π D. 55π

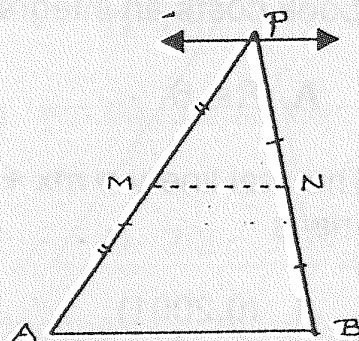


28. Points M and N are the mid points of sides PA and PB of triangle PAB. As P moves along a line that is parallel to side AB, consider the following:

- I. The length MN
- II. The area of triangle PAB
- III. The area of trapezoid ABNM
- IV. The perimeter of triangle PAB

Which of the above will change?

- A. I & II only B. II & III only C. II only D. IV only



29. In the given figure ABCD is a quadrilateral whose diagonals meet at O. If ΔAOB , ΔDOC and ΔBOC have areas 3, 10, and 2 respectively then the area of ΔAOD is

- A. $\sqrt{30}$ B. $\sqrt{15}$ C. $\sqrt{60}$ D. $\sqrt{20}$

